0. **Code:** bhagams4063 bhagamsaigoutham exam.zip

1. exam/exam/Program.cs

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```
using System.Reflection.Metadata.Ecma335;

namespace exam
{
    internal class Program
    {
        static void Main(string[] args)
        {
            int[] coeffi1 = {
                5,
                6,
                7,
                8,
                9
            };
            int n1 = 2;
            int[] result = polynomial(coeffi1, n1);
            Console.WriteLine(string.Join(",", result));
        }
        static int[] polynomial(int[] coeffi, int n) //creating two parameters in a method
        {
            if (coeffi == null || coeffi.Length == 0 || n ≤ 0) //null values
            {
                return null;
            }

            int resulting_length = n * (coeffi.Length - 1) + 1;
            int[] result = new int[resulting_length]; //creating a method that returns resulting length
```

B1

B2

1. **exam/exam/Program.cs**

```
            //copy the orginal coefficients to end of the result array

            for (int i = 0; i < coeffi.Length; i++)
            {
                result[result.Length - coeffi.Length + i] = coeffi[i];
            }
            //polynomial multiplication
            for (int i = 1; i < n; i++)
            {
                for (int j = result.Length - 1; j ≥ 0; j--)
                {
                    int temp = 0;
                    for (int k = 0; k < coeffi.Length; k++)
                    {
                        temp = result[j] * coeffi[k];
                    }
                    result[j] = temp;
                }

            }


            return result;

        }

    }
}
```

B3

sum

int[]sum = new int[result.length];

B6

if(result[j] ==0) continue;

B5

sum[j - (coeffi.Length - 1 - k)] += must be shifted!

B4

result = sum;

needs to create a copy then set to result.

×1

B3